

CSC258 Assembly Project : Breakout

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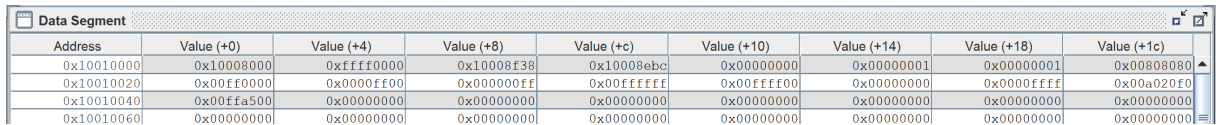
1 About the Game

1. Explain your plan for what will be stored in memory, and how it will be stored and laid out.

The following information needs to be stored in the memory:

Label Name	Size (bytes)	Description
ADDR_DSPL	4	Address of the Display
ADDR_KBRD	4	Address of the Keyboard
ADDR_PADDLE	4	Initial Address of the Paddle
ADDR_BALL	4	Initial Address of the Ball
ADDR_BRICK	4	Address of the Falling brick
BALL_VEL_X	4	X- component of Ball Velocity
BALL_VEL_Y	4	Y- component of Ball Velocity
WALL	4	Color of the Wall
RED	4	Color of first layer of bricks
BLUE	4	Color of second layer of bricks
GREEN	4	Color of third layer of bricks
WHITE	4	Color of paddle and unbreakable bricks
YELLOW	4	Color of ball
BLACK	4	Color of empty regions
ORANGE	4	Color of brick when 3 hits left for it to be broken.
PURPLE	4	Color of Brick when it falls
CYAN	4	Color of brick when 2 hits left for it to be broken.

2. Show that your memory has been laid out according to your plan by including a screenshot (or multiple screenshots) of memory.



Address	Value (+0)	Value (+4)	Value (+8)	Value (+c)	Value (+10)	Value (+14)	Value (+18)	Value (+1c)
0x10010000	0x10008000	0xfffff000	0x10008f38	0x10008ebc	0x00000000	0x00000001	0x00000001	0x00808080
0x10010020	0x00ff0000	0x0000ff00	0x000000ff	0x00ffffff	0x00fffff0	0x00000000	0x0000ffff	0x00a020f0
0x10010040	0x00ffa500	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010060	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000

Figure 1: Variables in Memory layout

[illegible]

Data Segment								
Address	Value (+0)	Value (+4)	Value (+8)	Value (+c)	Value (+10)	Value (+14)	Value (+18)	Value (+1c)
0x10008200	0x00080808	0x000000ff	0x000000ff	0x000000ff	0x000000ff	0x000000ff	0x000000ff	0x000000ff
0x10008220	0x000000ff	0x000000ff	0x000000ff	0x000000ff	0x00ffffff	0x000000ff	0x000000ff	0x000000ff
0x10008240	0x000000ff	0x000000ff	0x000000ff	0x00ffffff	0x000000ff	0x000000ff	0x000000ff	0x000000ff
0x10008260	0x000000ff	0x000000ff	0x000000ff	0x000000ff	0x000000ff	0x000000ff	0x000000ff	0x00080808
0x10008280	0x00080808	0x0000ff00	0x0000ff00	0x0000ff00	0x0000ff00	0x0000ff00	0x0000ff00	0x0000ff00
0x100082a0	0x0000ff00	0x0000ff00	0x00ffffff	0x0000ff00	0x0000ff00	0x0000ff00	0x0000ff00	0x0000ff00
0x100082c0	0x0000ff00	0x0000ff00	0x0000ff00	0x0000ff00	0x0000ff00	0x00ffffff	0x0000ff00	0x0000ff00
0x100082e0	0x0000ff00	0x0000ff00	0x0000ff00	0x0000ff00	0x0000ff00	0x0000ff00	0x0000ff00	0x00080808
0x10008300	0x00080808	0x00ff0000	0x00ff0000	0x00ff0000	0x00ff0000	0x00ff0000	0x00ff0000	0x00ff0000
0x10008320	0x00ffffff	0x00ff0000	0x00ff0000	0x00ff0000	0x00ff0000	0x00ff0000	0x00ff0000	0x00ff0000
0x10008340	0x00ff0000	0x00ff0000	0x00ff0000	0x00ff0000	0x00ff0000	0x00ff0000	0x00ff0000	0x00ffffff
0x10008360	0x00ff0000	0x00ff0000	0x00ff0000	0x00ff0000	0x00ff0000	0x00ff0000	0x00ff0000	0x00080808
0x10008380	0x00080808	0x000000ff	0x000000ff	0x000000ff	0x00ffa500	0x000000ff	0x00ffffff	0x00ffffff
0x100083a0	0x000000ff	0x000000ff	0x000000ff	0x0000ffff	0x000000ff	0x000000ff	0x000000ff	0x000000ff
0x100083c0	0x000000ff	0x000000ff	0x000000ff	0x000000ff	0x000000ff	0x000000ff	0x000000ff	0x000000ff
0x100083e0	0x00ffffff	0x000000ff	0x000000ff	0x00ffa500	0x000000ff	0x000000ff	0x000000ff	0x00080808

◀ | ▶
current \$gp ▾
☒ Hexadecimal Addresses
☒ Hexadecimal Values
☐ ASCII

Figure 3: Brick units in Memory

[illegible]

Figure 4: Initial Position of Paddle units in Memory start at the address stored in ADDR_PADDLE Label

[illegible]

Figure 5: Initial Position of Ball in Memory

3. Show the static scene explained in Milestone 1.

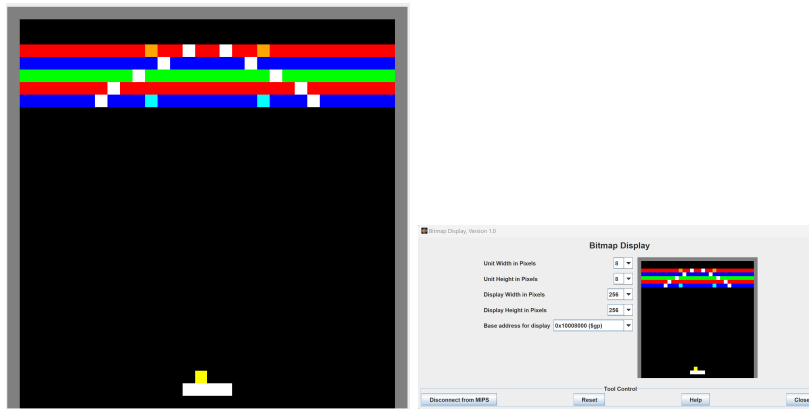


Figure 6: Static Scene

4. Explain how the ball change directions when it collides.

The x and y components of velocities are maintained in `BALL_VEL_X` (possible values are 0, 1 and -1) and `BALL_VEL_Y` (possible values are -1 and 1), and updated after each collision. When there is movement, it checks first along the x direction if the the next position of the ball coincides with the position of a wall or a brick. If it does, then there is a collision along the x component of the ball, and the x component of velocity of the ball is inverted. A similar method is used to detect for collision along y axis and update y component of velocity. The new ball drawn is using the new velocity vectors.

2 How To Play: Instructions for the Game

1. To start the game, press the 's' key.
2. To move the paddle, use key 'a' to move the paddle to 'left' and use key 'd' to move the paddle to 'right'.
3. The paddle cannot be moved 'up' or 'down'.
4. To pause the game, press 'p' key.
5. If the ball goes below the paddle, the game is lost.
6. The white bricks in the game are unbreakable.
7. The orange bricks need to be hit 3 times to break. When hit once, they change to cyan bricks.
8. The cyan bricks need to be hit 2 times to break. When hit once, they change to green bricks.
9. To quit the game, press 'q' key.
10. Have Fun !!